# Detecting Deepfake Audio Through Physical Acoustic Features

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## Background

- Modern TTS and voice cloning systems can generate highly convincing synthetic speech, posing significant societal and cybersecurity risks.
- Traditional detectors often rely on model-specific spectral artifacts, limiting generalization across unseen data (e.g. new models, accents, languages).
- Authentic speech is naturally constrained by physical processes, including vocal fold excitation, vocal tract anatomy, and the recording environment's acoustics, that leave measurable signatures.
- TTS systems mimic perceptual sound, rather than physical plausibility, so they struggle to accurately reproduce these features.

## Research Question

How can we reliably detect deepfake audio by using physical acoustic features derived from speech production and the recording environment?

## Hypothesis

- We predict that synthetic speech fails to naturally reproduce the physical signatures of genuine speech production and capture.
- Consequently, we predict that jointly modeling **glottal excitation (GP)**, **vocal tract geometry (VT)**, and **the recording environment (EA)** will further improve deepfake detection performance across diverse synthesis models instead of using a singular model.

## Methods

- Constructed corpus with real & fake speech recordings from publicly available datasets and TTS models.* Standardized audio and created speaker-disjoint train/validation/test splits to prevent speaker leakage.

Applied LPC inverse filtering to extract raw glottal waveforms, generated 128-band Mel-spectrograms with 1K samples; **2D CNN classifier**

Modeled vocal tract as a tube resonator, used LPC reflection coefficients to derive cross-sectional articulatory tube areas; **Random Forest classifier**

Extracted 35 room features capturing reverberation (decay-slope RT, DRR), noise-floor & channel stats, digital silence; **Gradient-boosted tree classifier**

- Evaluated performance on separate test sets using balanced accuracy, ROC-AUC, and equal error rate (EER).
- Repeated pipeline across 10 random seeds, performed feature ablations to quantify individual contribution.

- United all three feature models in a single ML pipeline, leveraging the outputs of the individual models as combined inputs for final classification

## Results

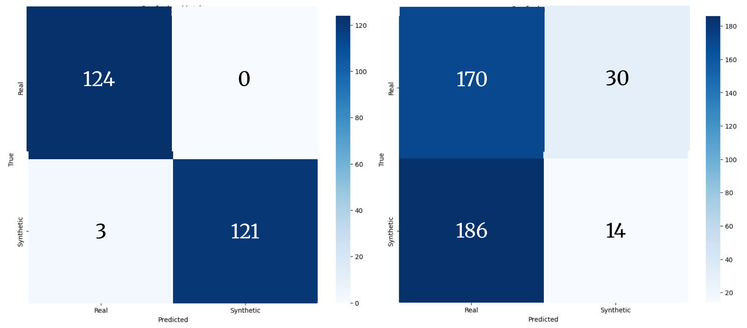



Figure 1: **GP** confusion matrices for testing held-out, speaker-disjoint set (left) vs. an unseen corpus (right).

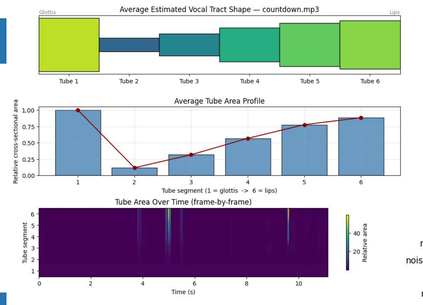



Figure 2: Visual representation of vocal tract cross-sectional area and change over time.

| | model | val_acc | val_bacc | val_auc | fit_secs |
|---|---|---|---|---|---|
| 0 | HistGradientBoosting | 0.9742 | 0.9743 | 0.9937 | 1.800000 |
| 1 | LightGBM | 0.9710 | 0.9711 | 0.9946 | 1.800000 |
| 2 | XGBoost | 0.9661 | 0.9663 | 0.9926 | 1.300000 |
| 3 | RandomForest | 0.9323 | 0.9324 | 0.9833 | 4.200000 |
| 4 | ExtraTrees | 0.9306 | 0.9308 | 0.9850 | 2.300000 |
| 5 | LogisticRegression | 0.8435 | 0.8439 | 0.9089 | 0.200000 |

→ selected model: HistGradientBoosting (val acc 0.9742)

Table 1: Model selection across classifier families for the **EA** branch.

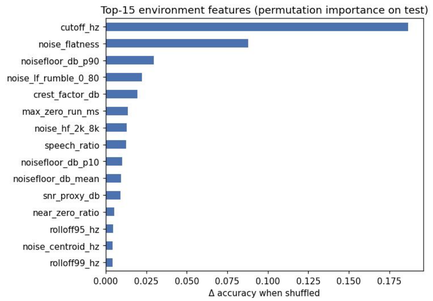



Figure 3: Bar plot of most influential **EA** features from permutation importance.

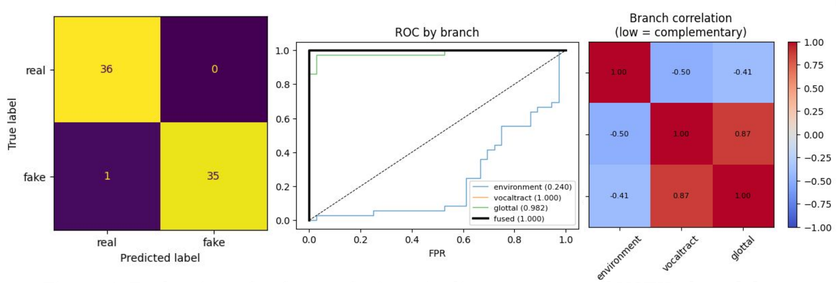



Figure 4: Evaluation of soft-vote fusion pipeline on our small VCTK-cloned dataset. The **GP** and **VT** models greatly outperform the environment model as they both were exposed to similar data during training.

## Discussion

- Individually, each model separates real from synthetic almost perfectly on the held-out, speaker-disjoint split for the same corpus, but accuracy collapses on an unseen corpus. Fixing this gap is the motivation for fusion.
- Figure 3 ranks the most influential **EA** features, supporting interpretable classification by displaying which signatures break down in deepfakes.
- Figure 4 shows **GP** and **VT** substantially outperforming **EA** on the VCTK-cloned set, which is expected given both saw similar data in training. The low inter-branch correlations are promising for fusion, but confirming this requires a fully held-out evaluation set, which is currently in progress.
- These results suggest a single model trained jointly on all three feature sets may outperform late fusion of three independently-trained branches.

## Applications

- Cybersecurity: Integrates directly into voice-lock systems, phone banking, and telecom to flag social engineering and voice-phishing attacks.
- Digital forensics: Provides legal professionals with an interpretable tool to verify the authenticity of audio evidence, rather than a "black-box" score
- Media & public safety: Enables social media platforms to flag synthetic speech / audio at upload, before false reports spread to consumers

## Future Work

- Expand data from across demographics, languages, dialects, and capturing modalities to improve broader application
- Deepen the physical feature set, across all three tested categories, including fine-grained representations that capture within-clip variation.
- Perform ablation studies to understand strongest features and synergy. For final output, test ensemble classifier, fine-tuning, and other fusion tactics.

## References

*Speech recordings sourced from LibriSpeech, VCTK, Gary Stafford Deepfake Audio Detection, 3004 Lakshu, and In-The-Wild; voice-cloning from xTTS-v2.


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